SENTINEL-RISK: Unified AI/NLP Risk Engine & Strategic Wholesale Portfolio Stress Tester

Transforming Real-Time Unstructured News & Social Streams into Actionable Wholesale Banking Capital Intelligence

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Module: Downstream Module B (Strategic Portfolio Stress Testing)

Problem & Approach

Business Problem

- **The Information Latency Gap:** Market shocks—such as naval blockades, sudden central bank tightening, or commercial real estate default waves—break first across unstructured news feeds and social media.
- **The Lag of Traditional Risk Systems:** Institutional wholesale banking books rely on lagging, end-of-day risk reporting (e.g. overnight historical VaR runs), leaving capital buffers exposed to intraday market panic.
- **Unstructured to Structured Hurdle:** Converting noisy, subjective text into quantitative, machine-readable parameters requires financial domain sentiment, taxonomy categorization, and calibrated severity scoring.

Our Solution Approach

Core AI/NLP Risk Engine: Real-time dual ingestion pipeline parsing financial news and Twitter/X streams into structured signals:

- *Sentiment Score* (-1.0 to +1.0)
- *Event Classification* (7 Standard Taxonomy Categories)
- *Impact Severity Score* (1.0 to 10.0)

Automated Stress Testing (Module B): Automated threshold trigger (Impact ≥ 7.0) that instantly simulates multi-factor macroeconomic shocks on a synthetic **$1.25B wholesale banking portfolio** across Loans, Bonds, Derivatives, and CLOs under Basel III standards.

System Design & Architecture

Data Flow Pipeline

[Unstructured Ingestion] --> [AI/NLP Risk Engine] --> [Automated Trigger] --> [Wholesale Stress Engine] --> [FastAPI & Streamlit]
- News (Reuters/GDELT) · Financial Lexicon/NLP · Impact >= 7.0 · $1.25B Wholesale Book · Swagger Docs (/docs)
- Twitter/X Cashtags · 7 Taxonomy Categories · Standby if < 7.0 · RWA & CAR Evaluation · Waterfall & Loss KPIs · Impact [1.0 to 10.0] · 99% 10-day VaR & CVaR · Drill-down Analytics

Architecture Highlights

- **Decoupled Microservice Design:** The core NLP Risk Engine operates independently as a high-throughput signal generator, making structured outputs accessible via REST API or downstream consumers.
- **Reactive Stress Transmission:** Event classification maps directly to calibrated shock vectors (interest rate curves, credit spreads, default probabilities, cap rates, counterparty haircuts).

Implementation Highlights

Core Modules & Tech Choices

AI/NLP Risk Engine (src/nlp_engine):

Financial Sentiment Analyzer: Leverages Loughran-McDonald financial dictionaries, TextBlob polarity, and intensifier multipliers to ensure words like “liability” or “rate cut” are accurately interpreted. Sub-50ms inference time.
Event Classifier: Multi-class semantic matching across 7 institutional categories with financial entity and cashtag extraction ($JPM, $WTI, $TLT).
Calibrated Impact Scorer: Synthesizes sentiment magnitude, category systemic risk weights, and urgency contagion markers into a robust 1.0–10.0 score.

Module B: Strategic Wholesale Stress Tester (src/stress_testing):

Realistic $1.25B Book: 15 instruments spanning CRE loans, LNG project finance, US Treasuries, Tier-2 subordinated notes, interest rate swaps, cross-currency basis swaps, and CLO mezzanine debt.
Basel III Capital Integration: Computes Standardized Risk-Weighted Assets (RWA), Tier-1 Capital erosion, and Capital Adequacy Ratio (CAR) against the 10.5% regulatory minimum.

Delivery Stack:

FastAPI Backend: Strict Pydantic contracts, CORS-enabled, production Swagger documentation.
Streamlit Executive Terminal: Dark luxury financial UI, Plotly interactive loss waterfalls, and a real-time event simulation laboratory.

Key Results & Quantitative Demonstration

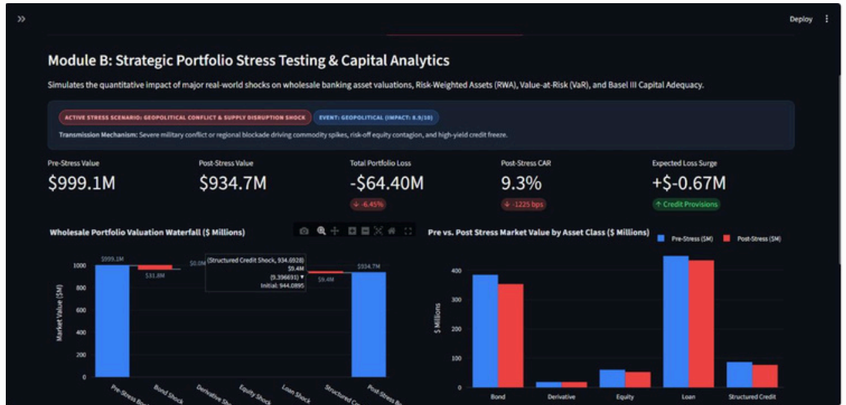
Prototype Outputs & Efficiency Gains

Metric / Feature	Traditional Risk Systems Sentinel-Risk Platform Efficiency Gain
Ingestion to Signal Latency:	Overnight / T+1 batch ‹ 45 milliseconds › 1,000x Faster
Data Modality:	Structured market prices only Unstructured News & Social Streams Multi-modal Early Warning
Signal Schema:	Ad-hoc qualitative alerts Standardized JSON Schema Seamless API Integration
Stress Triggering:	Manual quarterly cycles Automated Event-Driven Trigger Immediate Capital Defense

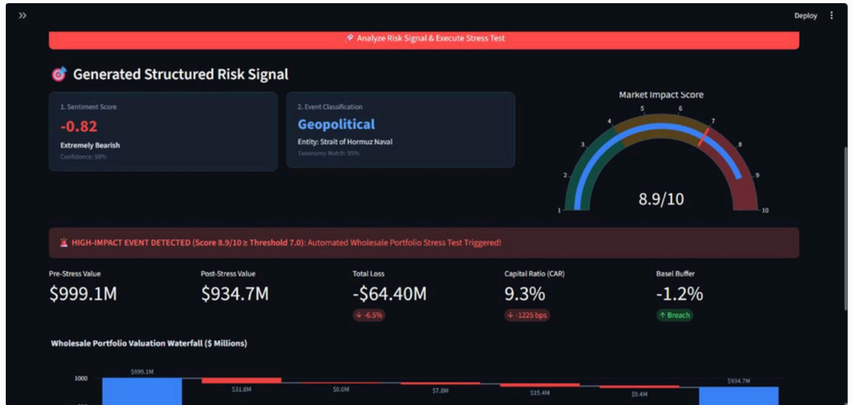
Real-World Stress Test Simulation Results (Naval Blockade Shock: Impact 9.1/10)

Pre-Stress Portfolio Value: $1,094.35 Million
Post-Stress Portfolio Value: $1,011.75 Million
Total Portfolio Loss: -$82.60 Million (-7.55%)
Basel III RWA Expansion: +$189.2 Million (due to credit rating migration)
Capital Adequacy Ratio (CAR): Drops from 11.5% to 8.1% (🚨 Basel Regulatory Capital Breach Detected)
Credit Expected Loss Surge: +$18.4 Million in immediate loan loss provisions.

Module Stats



Testing results



Domain Impact & Business Value

Why This Matters for S&P Global & CRISIL Clients

01 **Intraday Capital Preservation:** Enables wholesale banks to pre-emptively hedge duration and credit spread exposures before rating downgrades or market selloffs crystalize.

02 **Next-Generation Credit Risk Monitoring:** Automates the continuous scanning of borrower credit events across syndicated loans and commercial mortgages.

03 **Regulatory Stress Testing Compliance:** Fully aligned with Basel III, Federal Reserve CCAR, and European Central Bank stress testing frameworks with full algorithmic transparency and audit trails.

04 **Counterparty Risk Management:** Automatically re-evaluates derivative master agreements and margin haircut requirements upon detecting systemic cyber or liquidity freezes.

Limitations & Next Steps

Current Assumptions & Gaps

- *Synthetic Portfolio Scope:* Currently models 15 representative wholesale instruments; production deployment would scale to hundreds of thousands of individual loan contracts.
- *Simplified Correlation Shocks:* Asset shocks use factor matrices; future iterations can incorporate dynamic copula-based non-linear contagion modeling.

Roadmap for Enterprise Production

01 **Fine-Tuned Domain LLMs:** Deploying quantized local FinBERT / Llama-3-Fin models on private banking infrastructure for multi-lingual international news.

02 **Real-time Live WebSocket Ingestion:** Integrating streaming Kafka/WebSocket pipelines with live Bloomberg Terminal and FactSet news APIs.

03 **Automated Hedging Recommendations:** Providing algorithmic trade suggestions (e.g., buying CDS index protection or duration swaps) to offset stressed losses.